

[25 marks]

Exercise I.

Compute the following integrals:

1. $\int \frac{1}{x(1-x)} dx,$
2. $\int \frac{1}{\sin x(\cos x - \sin x)} dx,$
3. $\int x^5 \ln x dx,$
4. $\int \frac{\sqrt[3]{x}}{\sqrt[4]{x^5} + \sqrt[4]{x^7}} dx,$
5. $\int (\arcsin x)^3 dx,$
6. $\int \frac{1}{x\sqrt{1-x^2}} dx.$

Exercise II.

Let $\alpha \in \mathbb{R}$. Study the natures of the following series:

1. $\sum_{n=1}^{\infty} (-1)^n \frac{1+e^{-n}}{\arctan n},$
2. $\sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1},$
3. $\sum_{n=1}^{\infty} \frac{3^{n+2} - 3n}{n3^{n+1}},$
4. $\sum_{n=1}^{\infty} \frac{\sin(\frac{1}{n})}{\sqrt{n} + n^2}.$
5. $\sum_{n=2}^{\infty} \frac{\ln n}{n} \alpha^n,$
6. $\sum_{n=1}^{\infty} \frac{e^n}{e^n + 1} \frac{(-1)^n}{n}.$

Exercise III.

Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals :

1. $\int_0^1 \frac{\sqrt{1-x}}{\sqrt{x-x}} dx,$
2. $\int_0^{\infty} \frac{\sin x}{1+x^2} dx,$
3. $\int_1^{\infty} \ln(1 + \frac{\sin x}{x^{\frac{2}{3}}}) dx,$
4. $\int_0^{\infty} \frac{(\sin x)^2}{x^{\alpha}} dx.$

Exercise IV.

1. Calculate the following limit:

$$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n e^{\sqrt{\frac{n-k}{n}}} \right)^{\frac{1}{n}}.$$

2. Let $g(x) = \int_{2x}^{\ln x} \ln(t-1) dt$. Compute $g'(x)$ and specify the domain of differentiability of g .

Exercise V.

Let f be a continuous function on $]0, \infty[$ such that $\lim_{x \rightarrow \infty} f(x) = \ell \in \mathbb{R}$ and $\int_0^1 \frac{f(t)}{t} dt$ is convergent.

1. Show that for $X > \varepsilon > 0$ we have

$$\int_{\varepsilon}^X \frac{f(2x) - f(3x)}{x} dx = \int_{2\varepsilon}^{3\varepsilon} \frac{f(t)}{t} dt - \int_{2X}^{3X} \frac{f(t)}{t} dt.$$

2. Show that

$$\lim_{X \rightarrow \infty} \int_{2X}^{3X} \frac{f(t)}{t} dt = \ell \ln \frac{3}{2}.$$

3. Show that $I = \int_0^{+\infty} \frac{f(2x) - f(3x)}{x} dx$ converges and give its value.

Ex1. Compute the following integrals:

$$1. \int \frac{1}{x(1-x)} dx = \int \frac{1}{x} dx + \int \frac{1}{1-x} dx;$$

$$= \ln|x| - \int \frac{1}{x-1} dx$$

$$\stackrel{(2)}{=} \ln|x| - \ln|x-1| + C.$$

$$\frac{1}{x(1-x)} \stackrel{(1)}{=} \frac{a}{x} + \frac{b}{1-x} \Rightarrow 1 = a(1-x) + bx$$

$$x=0 \Rightarrow a=1 \quad \& \quad x=1 \Rightarrow b=1$$

$$2. \int \frac{1}{\sin x (\cos x - \sin x)} dx = \int \frac{1}{\sin x (\cos x - \sin x)} \cdot \frac{1}{\cos^2 x} dx;$$

$$\stackrel{(1)}{=} \int \frac{1}{t(1-t)} dt \stackrel{(1)}{=} \ln|t| - \ln|t-1| \Big|_{t=\tan x} + C = \ln|\tan x| - \ln|\tan x - 1| + C$$

$$\tan x = t, \quad \frac{1}{\cos^2 x} dx = dt$$

$$3. \int x^5 \ln x dx \stackrel{(2)}{=} \frac{x^6}{6} \ln x - \frac{1}{6} \int x^5 dx + C = \frac{x^6}{6} \ln x - \frac{1}{36} x^6 + C$$

D	P
$\ln x$	x^5
$\frac{1}{x}$	$\frac{x^6}{6}$

$$4. \int \frac{\sqrt[3]{x}}{\sqrt[4]{x^3} + \sqrt[4]{x^7}} dx = \int \frac{x^{\frac{1}{3}}}{x^{\frac{3}{4}} + x^{\frac{7}{4}}} dx;$$

$$\stackrel{(2)}{=} \int \frac{t^2}{t^3 + t^7} 4t^3 dt$$

$$\text{L.C.M}\{2, 4\} = 4$$

$$x = t^4, \quad dx = 4t^3 dt.$$

$$\stackrel{(1)}{=} 4 \int \frac{1}{1+t^4} dt \stackrel{(1)}{=} 4 \arctan t \Big|_{t=x^{\frac{1}{4}}} + C = 4 \arctan x^{\frac{1}{4}} + C.$$

$$5. \int (\arcsin x)^3 dx \stackrel{(1)}{=} \int t^3 \cos t dt$$

$$\arcsin x = t$$

$$\Rightarrow x = \sin t \wedge dx = \cos t dt$$

$$\stackrel{(1)}{=} t^3 \sin t - 3t^2 \cos t - 6t \sin t - 6 \cos t \Big|_{t=\arcsin x} + C.$$

D	P
t^3	$\cos t$
$3t^2$	$\sin t$
$6t$	$-\cos t$
6	$-\sin t$
0	$\cos t$

$$6. \int \frac{1}{x\sqrt{1-x^2}} dx = \int \frac{\cancel{\cos t} dt}{\sin t \sqrt{1-\sin^2 t}} \stackrel{(1)}{=} \int \frac{dt}{\sin t};$$

$$= \int \frac{\sin t dt}{\sin^2 t} \stackrel{(1)}{=} \int \frac{du}{1-u^2}$$

$$x = \sin t, \quad dx = \cos t dt$$

$$t = \arcsin x; \quad t \in [0, \frac{\pi}{2}]$$

$$\cos t = u$$

$$\Rightarrow \sin t dt = du$$

$$= - \int \frac{du}{u^2-1} \stackrel{(1)}{=} - \frac{1}{2} \ln \left| \frac{u-1}{u+1} \right| \Big|_{u=\arccos(\arcsin x)}$$

$$\int \frac{1}{x\sqrt{1-x^2}} dx = \int \frac{1}{x\sqrt{1-(x-1)(x+1)}} dx = \int \frac{1}{x(x+1)\sqrt{\frac{1-x}{1+x}}} dx;$$

$$\stackrel{(1)}{=} \int \frac{1}{\frac{1-t^2}{4t^2} \cdot \frac{2}{1+t} \cdot \frac{-4t}{(1+t)^2}} dt \stackrel{(1)}{=} 2 \int \frac{1}{1-t^2} dt$$

$$\stackrel{(1)}{=} \ln \left| \frac{t-1}{t+1} \right| \Big|_{t=\sqrt{\frac{1-x}{1+x}}}$$

$$\frac{1-x}{1+x} = t^2 \Rightarrow 1-x = t^2 + xt^2$$

$$\Rightarrow \frac{1-t^2}{1+t^2} = x$$

$$\wedge \frac{-4t}{(1+t^2)^2} dt = dx$$

Ex II. Let $\alpha \in \mathbb{R}$. Study the nature of the following series:

1. $\sum_{n=1}^{\infty} (-1)^n \frac{1+e^{-n}}{n \ln n}$; $\lim_{n \rightarrow \infty} |a_n| = \frac{1}{\ln 2} \neq 0 \Rightarrow$ Series div. (4)

2. $\sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1}$ (2) $= \sum_{n=2}^{\infty} \frac{\ln n}{n} = \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^1}$; $\alpha=1, \beta=-1 \leq 1 \Rightarrow$ Series div. (1)
 ("Bertrand")

or, Integral test: $f(x) = \frac{\ln(x+1)}{x+1}$ (1) $\Rightarrow f'(x) = \frac{1 - \ln(x+1)}{(x+1)^2} < 0$ near $+\infty$ for $n > 2 > e-1$

$1 - \ln(x+1) < 0 \Leftrightarrow 1 < \ln(x+1)$
 $\Leftrightarrow e < x+1$
 $\Leftrightarrow x > e-1$

f cont. $\left(\frac{\ln(x+1)}{x+1}\right)$, positive, decreasing on $[2, +\infty]$ $\Rightarrow \sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1}$ and $\int_2^{\infty} \frac{\ln(x+1)}{x+1} dx$ have same nature.

$\int_2^{\infty} \frac{\ln(x+1)}{x+1} dx = \int_2^{\infty} \ln(x+1) d(\ln(x+1)) = [\ln(x+1)]^2 \Big|_2^{\infty} = \infty$ div $\Rightarrow \sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1}$ div. (2)

3. $\sum_{n=1}^{\infty} \frac{3^{n+2} - 3n}{n 3^{n+1}}$ (1) $= \sum_{n=1}^{\infty} \left[3 \cdot \frac{1}{n} - \frac{1}{3^n} \right]$ (1) div (bec $\sum \frac{1}{n}$ div (Riem) & $\sum (\frac{1}{3})^n$ conv (G.S. $|\frac{1}{3}| < 1$))

4. $\sum_{n=1}^{\infty} \frac{\sin(1/n)}{\sqrt{n} + n^2}$; $\frac{\sin \frac{1}{n}}{\sqrt{n} + n^2} \sim \frac{1}{n^2} = \frac{1}{n^2} > 0$; $\sum \frac{1}{n^2}$ conv (Riem) $\Rightarrow$ Series conv. (1)

5. $\sum_{n=2}^{\infty} \frac{\ln n}{n} \alpha^n$; Ratio test: $\left| \frac{U_{n+1}}{U_n} \right| = \left| \frac{\ln(n+1) \alpha^{n+1}}{n+1} \cdot \frac{n}{\ln n \alpha^n} \right| = \frac{n}{n+1} \frac{\ln(n+1)}{\ln n} |\alpha| \xrightarrow{n \rightarrow \infty} |\alpha|$ (1)

• $|\alpha| < 1 \Rightarrow$ Series conv absol. • $|\alpha| > 1 \Rightarrow$ Series div.

• $|\alpha| = 1 \Leftrightarrow \alpha = \pm 1$: $\alpha = 1, \sum \frac{\ln n}{n}$ div Bertrand. (1)

$\alpha = -1, \sum \frac{\ln n}{n} (-1)^n$; $b_n = \frac{\ln n}{n} > 0, b_n \xrightarrow{n \rightarrow \infty} 0$ conv, A.L.S.T. (1)

6. $\sum_{n=1}^{\infty} \frac{e^n}{e^{n+1}} \frac{(-1)^n}{n}$; ABEL'S test (1)
 convergent

• $\sum \frac{(-1)^n}{n}$ conv (A.L.S.T. $\frac{1}{n} > 0, \frac{1}{n} \xrightarrow{n \rightarrow \infty} 0$) (1)

• $b_n = \frac{e^n}{e^{n+1}}$ bdd ($0 < b_n < 1$), monotone (1)

$f(x) = \frac{e^x}{e^{x+1}}, f'(x) = \frac{e^x}{(e^{x+1})^2} > 0$. (1B)

Ex III. Let $\alpha \in \mathbb{R}$. Study the nature of the following improper int:

1. $\int_0^1 \frac{\sqrt{1-x}}{\sqrt{x}-x} dx = \int_0^{1/2} \frac{\sqrt{1-x}}{\sqrt{x}-x} dx + \int_{1/2}^1 \frac{\sqrt{1-x}}{\sqrt{x}-x} dx = I_0 + I_1$ (1)

near 0^+ : $\frac{\sqrt{1-x}}{\sqrt{x}-x} \sim \frac{1}{x^{1/2}} > 0$, I_0 conv (equiv test + Riem $\alpha = 1/2 < 1$) (1)

near 1^- : $\frac{\sqrt{1-x}}{\sqrt{x}-x} = \frac{\sqrt{1-x}}{\sqrt{x}-x} \cdot \frac{\sqrt{x}+x}{\sqrt{x}+x} = \frac{\sqrt{1-x}(\sqrt{x}+x)}{x-x^2} = \frac{\sqrt{1-x}(\sqrt{x}+x)}{x(1-x)} \sim 2 \cdot \frac{1}{(1-x)^{1/2}} > 0$ (1)

I_1 conv (equiv test + Riem $\alpha = 1/2 < 1$) (1B)

Thus the imp int 1. conv.

2. $\int_0^{\infty} \frac{\sin x}{1+x^2} dx$

near ∞ : (1B) $0 < \left| \frac{\sin x}{1+x^2} \right| = \frac{|\sin x|}{1+x^2} \leq \frac{1}{x^2}$; $\int_1^{\infty} \frac{1}{x^2} dx$ conv (1) $\xrightarrow[\text{Riem}]{\text{Comparison test}} 2. \text{ Absol conv (Hence conv).}$

$$3. \int_1^{\infty} \ln\left(1 + \frac{\sin x}{x^{2/3}}\right) dx$$

near ∞ ; $\ln\left(1 + \frac{\sin x}{x^{2/3}}\right) \stackrel{(1)}{=} \frac{\sin x}{x^{2/3}} - \frac{1}{2} \frac{\sin^2 x}{x^{4/3}} + o\left(\frac{\sin^2 x}{x^{4/3}}\right)$; $\frac{\sin x}{x^{2/3}} \xrightarrow{x \rightarrow \infty} 0$

• $\int_1^{\infty} \frac{1}{x^{2/3}} \sin x dx$ Conv, Abel-Dir test | $g(x) = \frac{1}{x^{2/3}} \xrightarrow{x \rightarrow \infty} 0$, monotone $\downarrow$.

• $\int_1^{\infty} \frac{\sin^2 x}{x^{4/3}} dx$ Absol Conv (compt) $\Rightarrow \int_1^{\infty} o\left(\frac{\sin^2 x}{x^{4/3}}\right) dx$ Absol Conv.
 $0 < \frac{\sin^2 x}{x^{4/3}} < \frac{1}{x^{4/3}}$; $\frac{1}{x^{4/3}} \xrightarrow{x \rightarrow \infty} 0$; $\frac{1}{x^{4/3}} \leq \frac{1}{x^{1/2}}$.

Thus 3. is conv as sum of conv imp integrals.

$$4. \int_0^{\infty} \frac{\sin^2 x}{x^{\alpha}} dx = \int_0^1 \frac{\sin^2 x}{x^{\alpha}} dx + \int_1^{\infty} \frac{\sin^2 x}{x^{\alpha}} dx = I_1 + I_2.$$

near 0: $\frac{\sin^2 x}{x^{\alpha}} \stackrel{(1)}{\sim} \frac{x^2}{x^{\alpha}} = \frac{1}{x^{\alpha-2}} > 0$, I_1 conv iff $\alpha-2 < 1$ iff $\alpha < 3$.

near ∞ : $\frac{\sin^2 x}{x^{\alpha}} \stackrel{(1)}{\sim} \frac{1}{x^{\alpha}}$ conv, comparison test $0 < \frac{\sin^2 x}{x^{\alpha}} < \frac{1}{x^{\alpha}}$
 $\Rightarrow \alpha < 1$ div = div + conv; $\frac{\sin^2 x}{x^{\alpha}} \stackrel{(1)}{\sim} \frac{1 - \cos 2x}{2x^{\alpha}}$
 $\int_1^{\infty} \frac{1}{x^{\alpha}} dx$ div & $\int_1^{\infty} \frac{1}{x^{\alpha}} \cos 2x dx$ conv Abel-Dir $\frac{1}{x^{\alpha}} \xrightarrow{x \rightarrow \infty} 0$ & $|F(x)| = \left| \int_1^x \cos 2t dt \right| \leq 2$.
 $\Rightarrow \alpha = 0$, $\int_0^{\infty} \sin^2 x dx = \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\infty} = \infty$ div
 $\Rightarrow \alpha < 0$, $\frac{1}{x^{\alpha}} \xrightarrow{x \rightarrow \infty} \infty \Rightarrow \frac{1}{x^{\alpha}} > 1$ near $\infty \Rightarrow \frac{\sin^2 x}{x^{\alpha}} \geq \sin^2 x > 0$ (div by comp test) $\stackrel{(1)}{\Rightarrow}$

Thus 4. conv iff I_1 conv $\wedge I_2$ conv iff $\alpha < 3 \wedge \alpha > 1$ iff $1 < \alpha < 3$. $\stackrel{(1B)}{\Rightarrow}$

Ex IV.

1. calculate the following lit:

$$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n e^{\sqrt{\frac{n-k}{n}}} \right)^{1/n} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \sqrt{\frac{n-k}{n}}} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \ln(e^{\sqrt{\frac{n-k}{n}}})}$$

$$= \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \sqrt{1 - \frac{k}{n}}} = e^{\int_0^1 \sqrt{1-x} dx}$$

$$\stackrel{(1)}{=} e^{\left[-\frac{2}{3} (1-x)^{3/2} \right]_0^1} = e^{-\frac{2}{3} [0 - 1]} \stackrel{(1)}{=} e^{2/3}$$

$f \in C([0,1]) \subseteq R([0,1])$

2. let $g(x) = \int_{2x}^{\ln x} \ln(t-1) dt$. compute $g'(x)$ and specify domain of def of g .

• $g'(x) \stackrel{(1)}{=} (\ln x)' \ln(\ln x - 1) - (2x)' \ln(2x - 1)$
 $\stackrel{(1)}{=} \frac{1}{x} \ln(\ln x - 1) - 2 \ln(2x - 1).$

$I = ?$ Domain of diff of g :

① $J =]1, +\infty[$ Domain of cont of $f(t) = \ln(t-1)$

② $u(n) = \ln n \wedge v(n) = 2n$ diff on $I_1 \cap I_2 =]0, \infty[\cap \mathbb{R} =]0, \infty[$

③ $u(n) \in j \Leftrightarrow \ln n > 1 \Leftrightarrow n > e \Leftrightarrow n \in I_3 =]e, \infty[$

④ $v(n) \in j \Leftrightarrow 2n > 1 \Leftrightarrow n > 1/2 \Leftrightarrow n \in I_4 =]1/2, \infty[$

⑤ $I = I_1 \cap I_2 \cap I_3 \cap I_4 =]0, \infty[\cap]e, \infty[\cap]1/2, \infty[=]e, \infty[$

Ex. let $f \in C([0, \infty[)$ s.t. $\lim_{n \rightarrow \infty} f(n) = l \in \mathbb{R}$ and $\int_0^1 \frac{f(t)}{t} dt$ is convergent.

1. S.T for $X > \varepsilon > 0$: $\int_{\varepsilon}^X \frac{f(2x) - f(3x)}{x} dx = \int_{2\varepsilon}^{3\varepsilon} \frac{f(t)}{t} dt - \int_{2X}^{3X} \frac{f(t)}{t} dt$

$$\int_{\varepsilon}^X \frac{f(2x) - f(3x)}{x} dx = \int_{\varepsilon}^X \frac{f(2x)}{x} dx - \int_{\varepsilon}^X \frac{f(3x)}{x} dx \quad (\text{both integrands } \in C([\varepsilon, X]) \subseteq R([\varepsilon, X]))$$

$$\boxed{x = \frac{t}{2} = v(t) \in C^1([2\varepsilon, 2X])} \quad \boxed{x = \frac{t}{3} = v(t) \in C^1([3\varepsilon, 3X])}$$

$$= \int_{2\varepsilon}^{2X} \frac{f(t)}{t} dt - \int_{3\varepsilon}^{3X} \frac{f(t)}{t} dt$$

$$= \int_{2\varepsilon}^{3\varepsilon} \frac{f(t)}{t} dt + \int_{3\varepsilon}^{2X} \frac{f(t)}{t} dt - \left[\int_{3\varepsilon}^{2X} \frac{f(t)}{t} dt + \int_{2X}^{3X} \frac{f(t)}{t} dt \right]$$

$$= \int_{2\varepsilon}^{3\varepsilon} \frac{f(t)}{t} dt - \int_{2X}^{3X} \frac{f(t)}{t} dt$$

2. S.T $\lim_{X \rightarrow \infty} \int_{2X}^{3X} f(t) \frac{1}{t} dt = l \ln \frac{3}{2}$

$$\int_{2X}^{3X} \frac{f(t)}{t} dt \stackrel{\text{MVT}}{=} f(c_X) \int_{2X}^{3X} \frac{1}{t} dt$$

$$= f(c_X) [\ln 3X - \ln 2X]$$

$$\stackrel{①}{=} f(c_X) \ln \frac{3}{2} \xrightarrow{X \rightarrow \infty} l \ln \frac{3}{2}$$

① $[2X, 3X] \subseteq]0, \infty[$
 ② $f \in C([2X, 3X])$
 ③ $g \in C([2X, 3X]) \subseteq R([2X, 3X])$
 ④ $g > 0$ on $[2X, 3X]$ (not sign)
 ⑤ $\exists 2X < c_X < 3X$ s.t. $X \rightarrow \infty, c_X \rightarrow \infty$

3. S.T $I = \int_0^{\infty} \frac{f(2x) - f(3x)}{x} dx$ converges & give its value.

$$I = \int_0^{\infty} \frac{f(2x) - f(3x)}{x} dx \stackrel{①}{=} \lim_{\varepsilon \rightarrow 0} \lim_{X \rightarrow \infty} \int_{\varepsilon}^X \frac{f(2x) - f(3x)}{x} dx$$

$$\stackrel{①}{=} \lim_{\varepsilon \rightarrow 0} \left(\int_{2\varepsilon}^{3\varepsilon} \frac{f(t)}{t} dt - \int_{2X}^{3X} \frac{f(t)}{t} dt \right)$$

$$= 0 - l \ln \frac{3}{2} \quad \left(\int_0^1 \frac{f(t)}{t} dt \text{ conv} \Rightarrow \int_{2\varepsilon}^{3\varepsilon} \frac{f(t)}{t} dt \xrightarrow{\varepsilon \rightarrow 0} \int_0^1 \frac{f(t)}{t} dt \right)$$

Other methods:

Ex I.

$$1. \int \frac{1}{x(1-x)} dx = - \int \frac{1}{x^2-x} dx \stackrel{(1)}{=} - \int \frac{1}{x^2-x+\frac{1}{4}-\frac{1}{4}} dx \stackrel{(2)}{=} - \int \frac{1}{(x-\frac{1}{2})^2 - (\frac{1}{2})^2} dx$$

$$\stackrel{(2)}{=} \begin{cases} -\frac{1}{2 \cdot \frac{1}{2}} \ln \left| \frac{(x-\frac{1}{2}) - \frac{1}{2}}{(x-\frac{1}{2}) + \frac{1}{2}} \right| + C = -\ln \left| \frac{x-1}{x} \right| + C. \\ \text{or} \\ \frac{1}{\frac{1}{2}} \operatorname{arctanh} \left(\frac{x-\frac{1}{2}}{\frac{1}{2}} \right) + C \end{cases}$$

$$2. \int \frac{1}{\sin x (\cos x - \sin x)} dx = \int \frac{\frac{2}{1+t^2} dt}{\frac{2t}{(1+t^2)} \left[\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} \right]} = \dots \quad t = \tan \frac{x}{2}$$

Ex II.

$$2. \sum_{n=1}^{\infty} \frac{\ln(n+1)}{n+1} \text{ equiv test: } \frac{\ln(n+1)}{n+1} \stackrel{(2)}{\sim} \frac{\ln n}{n} > 0; \text{ But-d div } \alpha=1, \beta=-1 \leq 1$$

only: $\alpha=1, \beta=-1 \leq 1$, series div But-d.

$$3. \sum_{n=1}^{\infty} \frac{3^{n+2} - 3n}{n 3^{n+1}} \quad \frac{3^{n+2} - 3n}{n 3^{n+1}} \stackrel{(2)}{\sim} \frac{3^{n+2}}{n 3^{n+1}} = 3 \cdot \frac{1}{n} > 0; \quad \sum \frac{1}{n} \stackrel{(1)}{\text{div}} \quad \text{Series div.}$$

4. Comparison test works

$$6. \sum_{n=1}^{\infty} \frac{e^n}{e^n+1} \frac{(1)^n}{n} \quad \text{A.L.S. test: } b_n = \frac{e^n}{n(e^n+1)} > 0, \downarrow, \frac{(1)}{n} \rightarrow 0.$$

$$\left[\frac{e^x}{(e^x+1)x} \right]' \stackrel{(1)}{=} \frac{e^x}{e^x+1} \cdot \frac{1}{x} \left[\frac{1}{e^x+1} - \frac{1}{x} \right] \stackrel{(1)}{<} 0$$

for $x > 1 \cdot (e^x+1 > x^2)$

Ex III.

2. ABEL-DIV ✓